Linear Secret Sharing
&
Access Control Matrix

Revisiting Shamir Scheme

- **Secret s**
- **Share i:** $y_i = s + a_1 x_i + a_2 x_i^2 + \cdots + a_{t-2} x_i^{t-2} + a_{t-1} x_i^{t-1}$
- **Vector interpretation: scalar product**

$$y_i = [1, x_i, x_i^2, \cdots, x_i^{t-2}, x_i^{t-1}] \bullet [s, a_1, a_2, \cdots, a_{t-2}, a_{t-1}]$$

- **Coefficients a_i are random → call them r_i**

$$y_i = [1, x_i, x_i^2, \cdots, x_i^{t-2}, x_i^{t-1}] \bullet [s, r_1, r_2, \cdots, r_{t-2}, r_{t-1}]$$

Shamir scheme in matrix form

$$Ax = b$$

- ➔ A ➔ matrix, n x t — GIVEN
- ➔ x ➔ vector, t — [secret, rand, rand,...]
- ➔ b ➔ vector, n — resulting shares

- ➔ Example: (3,4)

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_3 & x_3^2 \\ 1 & x_4 & x_4^2 \end{pmatrix} \begin{pmatrix} s \\ r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

Special type of matrix
Vandermonde

Reconstructing secret

- **Linear system**
  - t entries
  - E.g. 3 shares out of 4

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_4 & x_4^2 \end{pmatrix} \begin{pmatrix} s \\ r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_4 \end{pmatrix}$$

Reconstruction coefficients:

- **Previous: Lagrange formula**
- **Now: MIGHT solve as linear system**
  - **Solution for s gives usual Laplace formula, of course**

$$\begin{pmatrix} s \\ r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_4 & x_4^2 \end{pmatrix}^{-1} \begin{pmatrix} y_1 \\ y_2 \\ y_4 \end{pmatrix}$$

Reconstructing secret

$$\begin{pmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ 1 & x_4 & x_4^2 \end{pmatrix}^{-1} = \begin{pmatrix} \frac{x_2 x_4}{(x_1 - x_2)(x_1 - x_4)} & \frac{x_1 x_4}{(x_2 - x_1)(x_2 - x_4)} & \frac{x_1 x_2}{(x_4 - x_1)(x_4 - x_2)} \\ \ldots & \ldots & \ldots \\ \ldots & \ldots & \ldots \end{pmatrix}$$

$$s = y_1 \frac{x_2 x_4}{(x_1 - x_2)(x_1 - x_4)} + y_2 \frac{x_1 x_4}{(x_2 - x_1)(x_2 - x_4)} + y_4 \frac{x_1 x_2}{(x_4 - x_1)(x_4 - x_2)}$$

An alternative interpretation

Row of party P_1	
Row of party P_2	
Row of party P_4	

$$\begin{pmatrix} \boxed{1 \quad x_1 \quad x_1^2} \\ \boxed{1 \quad x_2 \quad x_2^2} \\ \boxed{1 \quad x_4 \quad x_4^2} \end{pmatrix} \boxed{\begin{pmatrix} s \\ \text{randomness} \end{pmatrix}} = \begin{pmatrix} \boxed{y_1} \\ \boxed{y_2} \\ \boxed{y_4} \end{pmatrix}$$

Each row = vector

v_1, v_2, v_3

Such vectors "span" a 3D space

$c_1 v_1 + c_2 v_2 + c_3 v_3$

$$c_1 \begin{pmatrix} 1 \\ x_1 \\ x_1^2 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ x_2 \\ x_2^2 \end{pmatrix} + c_3 \begin{pmatrix} 1 \\ x_4 \\ x_4^2 \end{pmatrix}$$

An alternative interpretation

- **Vector [1,0,0] included in "span"**
  - Exists linear combination
    $c_1\mathbf{v_1}+c_2\mathbf{v_2}+c_3\mathbf{v_3}=[1,0,0]$

$$c_1\begin{pmatrix}1 & x_1 & x_1^2\end{pmatrix}+c_2\begin{pmatrix}1 & x_2 & x_2^2\end{pmatrix}+c_3\begin{pmatrix}1 & x_4 & x_4^2\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\end{pmatrix}$$

- **Found by solving different linear system**
  - **Result… guess what ☺**

$$\begin{pmatrix}1 & 1 & 1\\ x_1 & x_2 & x_4\\ x_1^2 & x_2^2 & x_4^2\end{pmatrix}\begin{pmatrix}c_1\\ c_2\\ c_3\end{pmatrix}=\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$$

$$c_1=\frac{x_2x_4}{(x_1-x_2)(x_1-x_4)};\quad c_2=\frac{x_1x_4}{(x_2-x_1)(x_2-x_4)};\quad c_3=\frac{x_1x_2}{(x_4-x_1)(x_4-x_2)}$$

An alternative interpretation

But…

$$\left\{c_1\begin{pmatrix}1 & x_1 & x_1^2\end{pmatrix}+c_2\begin{pmatrix}1 & x_2 & x_2^2\end{pmatrix}+c_3\begin{pmatrix}1 & x_4 & x_4^2\end{pmatrix}\right\}\begin{pmatrix}s\\ r_1\\ r_2\end{pmatrix}=\begin{pmatrix}1 & 0 & 0\end{pmatrix}\begin{pmatrix}s\\ r_1\\ r_2\end{pmatrix}=s$$

$$c_1\begin{pmatrix}1 & x_1 & x_1^2\end{pmatrix}\begin{pmatrix}s\\ r_1\\ r_2\end{pmatrix}+c_2\begin{pmatrix}1 & x_2 & x_2^2\end{pmatrix}\begin{pmatrix}s\\ r_1\\ r_2\end{pmatrix}+c_3\begin{pmatrix}1 & x_4 & x_4^2\end{pmatrix}\begin{pmatrix}s\\ r_1\\ r_2\end{pmatrix}=s$$

share

$$c_1y_1+c_2y_2+c_3y_4=s$$

SUMMARY: shamir secret sharing treated as "span" problem

Much more general!

- **Linear Secret Sharing Scheme (LSSS)**
  - Matrix can be ARBITRARY!

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{pmatrix} \begin{pmatrix} s \\ r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{pmatrix}$$

- **Equivalence with (monotone) “span programs”**
  - **Amos Beimel theorem: LSSS = MSP**

Trivial Secret Share is LSSS

➔ **Example: (3,3)**

$$\begin{pmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} s \\ r_1 \\ r_2 \end{pmatrix} = \begin{pmatrix} s - r_1 - r_2 \\ r_1 \\ r_2 \end{pmatrix}$$

➔ **Span program**

$$c_1(1 \quad -1 \quad -1) + c_2(0 \quad 1 \quad 0) + c_3(0 \quad 0 \quad 1) = (1 \quad 0 \quad 0)$$

$$c_1 = c_2 = c_3 = 1$$

➔ **Share rec.**

$$c_1(s - r_1 - r_2) + c_2 r_1 + c_3 r_2 = (s - r_1 - r_2) + r_1 + r_2 = s$$

Any LSSS is homomorphic

$$x_a = (s_a, r_{1a}, r_{2a}, \cdots) \quad y_a = (share_{1a}, share_{2a}, \cdots)$$

$$x_b = (s_b, r_{1b}, r_{2b}, \cdots) \quad y_b = (share_{1b}, share_{2b}, \cdots)$$

$$Ax_a = y_a$$

$$Ax_b = y_b$$

$$y_a + y_b = Ax_a + Ax_b = A(x_a + x_b) = A(s_a + s_b, rand, rand, \cdots)$$

Monotone Span Programs
(example in GF2, other fields OK)

P_2	1	0	1	1
P_2	0	1	1	0
P_1	0	1	1	0
P_3	0	0	1	1
P_4	1	1	0	0
	0	0	0	1

The program accepts a set *B*
iff
the rows labeled by *B* span the target vector.

Monotone Span Programs

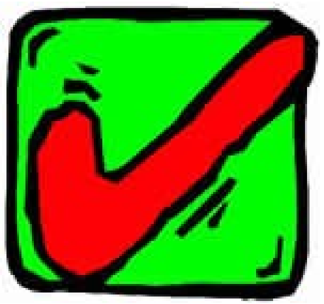
P_2	**1**	**0**	**1**	**1**
P_2	**0**	**1**	**1**	**0**
P_1	0	1	1	0
P_3	0	0	1	1
P_4	**1**	**1**	**0**	**0**
	0	0	0	1

1	**0**	**1**	**1**

$\oplus$

0	**0**	**1**	**1**
0	0	0	1

{P2,P4}



Slides from A. Beimel

Monotone Span Programs

P_2	1	0	1	1
P_2	0	1	1	0
P_1	0	1	1	0
P_3	0	0	1	1
P_4	1	1	0	0
	0	0	0	1

1	0	1	1
0	1	1	0
0	1	1	0

0 0 0 1

{P1,P2}

Slides from A. Beimel

Span Programs Secret Sharing

$$
\begin{array}{c}
P_2 \\ P_2 \\ P_1 \\ P_3 \\ P_4
\end{array}
\begin{bmatrix}
1 & 0 & 1 & 1 \\
0 & 1 & 1 & 0 \\
0 & 1 & 1 & 0 \\
0 & 0 & 1 & 1 \\
1 & 1 & 0 & 0
\end{bmatrix}
\begin{bmatrix}
s \\ r_2 \\ r_3 \\ r_4
\end{bmatrix}
=
\begin{bmatrix}
s + r_2 + r_4 \\ r_2 + r_3 \\ r_2 + r_3 \\ s + r_2 \\ r_3 + r_4
\end{bmatrix}
\begin{array}{c}
\mathrm{P2} \\ \mathrm{P2} \\ \mathrm{P1} \\ \mathrm{P3} \\ \mathrm{P}^4
\end{array}
$$

Example s=1,r2=r3=0, r4=1

$$
\begin{bmatrix}
0 \\ 0 \\ 0 \\ 1 \\ 1
\end{bmatrix}
\begin{array}{c}
\mathrm{P2} \\ \mathrm{P2} \\ \mathrm{P1} \\ \mathrm{P3} \\ \mathrm{P}^4
\end{array}
$$

Span Programs □ Secret Sharing

$$
\begin{array}{c}
P_2 \\ P_2 \\ P_1 \\ P_3 \\ P_4
\end{array}
\begin{bmatrix}
1 & 0 & 1 & 1 \\
0 & 1 & 1 & 0 \\
0 & 1 & 1 & 0 \\
0 & 0 & 1 & 1 \\
1 & 1 & 0 & 0
\end{bmatrix}
\begin{bmatrix}
s \\ r_2 \\ r_3 \\ r_4
\end{bmatrix}
=
\begin{bmatrix}
s+r_2+r_4 \\ r_2+r_3 \\ r_2+r_3 \\ s+r_2 \\ r_3+r_4
\end{bmatrix}
\begin{array}{c}
\text{P2} \\ \text{P2} \\ \text{P1} \\ \text{P3} \\ \text{P}^4
\end{array}
$$

1	0	0	0

s

{P2,P4}

LSSS and access structure

- **Every (monotone) access structure can be realized**

$$\text{participants}: P = \{P_1, P_2, P_3, P_4\}$$

$$\text{monotone access structure}: A \subseteq 2^P$$

$$example: A = \{\{P_1, P_2\}, \{P_1, P_3, P_4\}\}$$

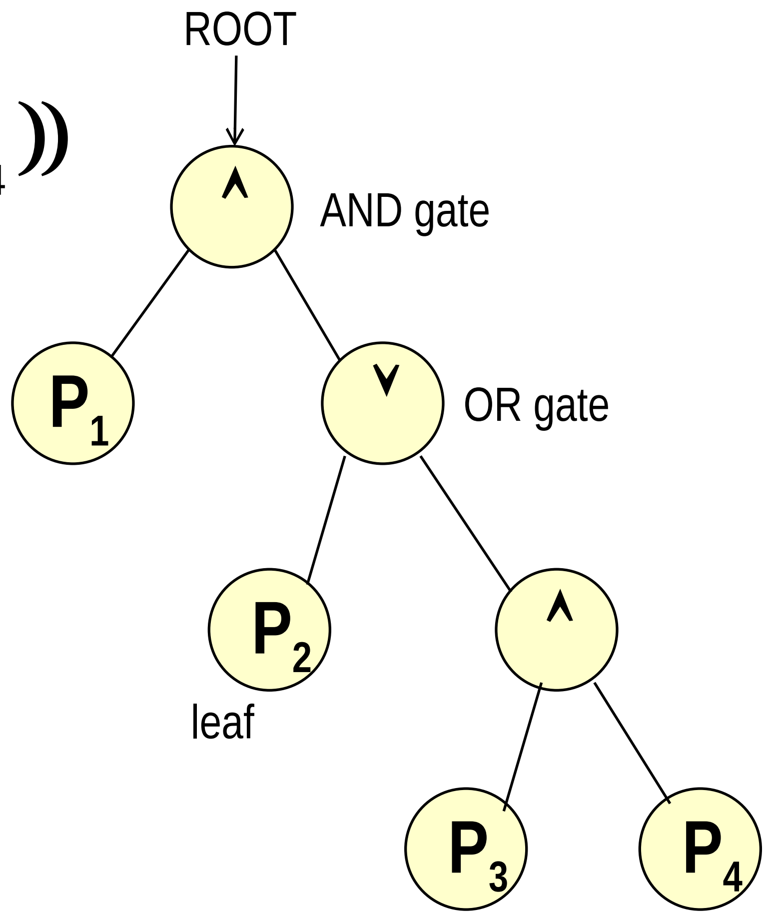
- **Consequence: every boolean predicate (without negation) may be supported**

$$example: P_1 \wedge (P_2 \vee (P_3 \wedge P_4))$$

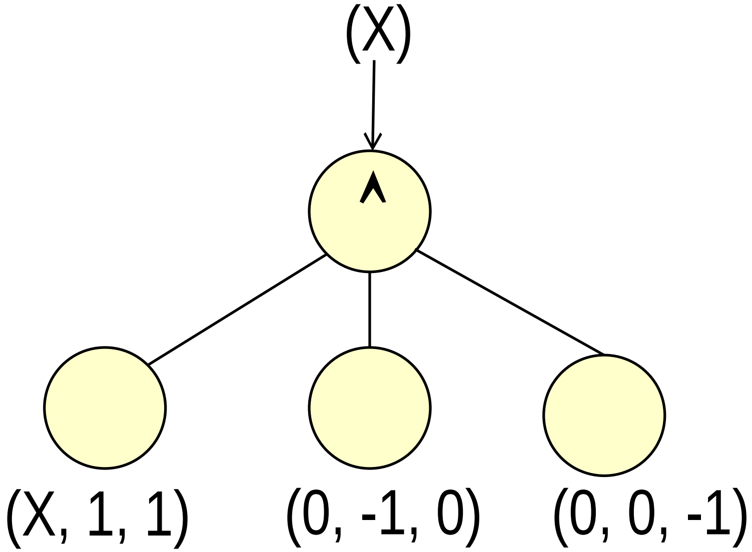
- **Scheme May NOT be ideal**
  - May entail more than 1 share per partner

LSSS matrix from AC Predicate

$$P_1 \wedge (P_2 \vee (P_3 \wedge P_4))$$

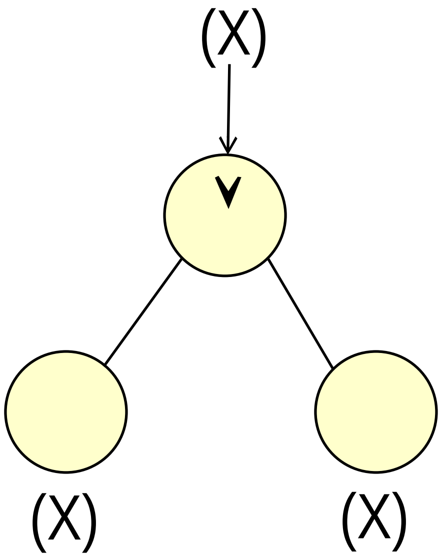


AND gate



$$\begin{pmatrix} X & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

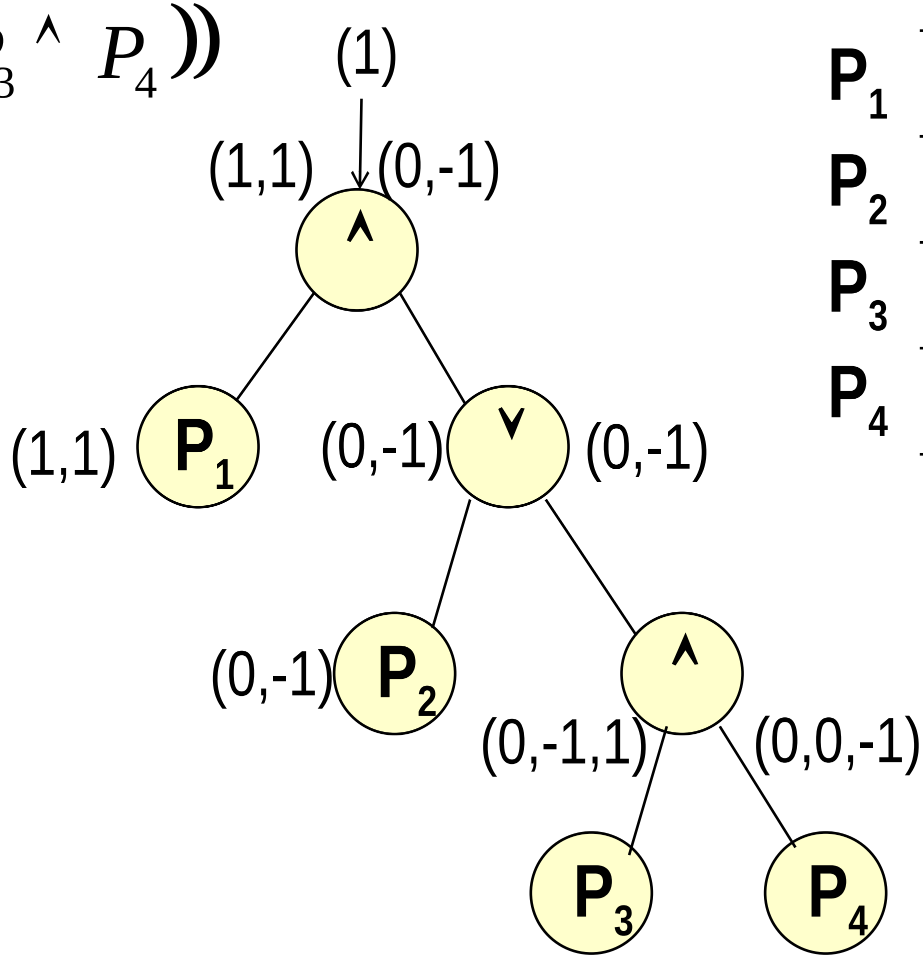
OR gate



$$\begin{pmatrix} X \\ X \end{pmatrix}$$

Matrix construction

$P_1 \wedge (P_2 \vee (P_3 \wedge P_4))$



P_1	1	1	0
P_2	0	-1	0
P_3	0	-1	1
P_4	0	0	-1

Careful with padding…

$$(A \wedge C \wedge D) \vee (B \wedge C)$$

$$\begin{pmatrix} A & 1 & -1 & -1 & . \\ C & 0 & 1 & 0 & . \\ D & 0 & 0 & 1 & . \\ B & 1 & . & . & -1 \\ C & 0 & . & . & 1 \end{pmatrix}$$

Wrap up

- **LSSS/MSP: generalization to arbitrary access structures**
  - Must be monotone
    - If parties A+B may access, also A+B+C may
    - Cannot model policies such as A+B+NOT(C)
      - Issue when revocation needed
- **Sub-optimal**
  - Parties may need more than 1 share
  - Minimum overhead: open research issue
- **Improved constructions**
  - Explicit threshold gates [Liu,Cao, 2010], but prime fields…
- **Applications**
  - Dramatic! Most modern crypto
    - E.g. attribute based cryptography